Algebraic Number Theory NMAL431
Homework 1 — Solutions

03 November 2026

Please send any comments and/or corrections to robin.visser@matfyz.cuni.cz

1. There are many possible answers one could give to this question. We provide some examples of suitable domains R below:

   (a) E.g. take $R = \mathbb{Z}[X]$ (or $\mathbb{Z}[X, Y]$ or $\mathbb{Z}[X_1, \ldots, X_n]$).

   - *R integrally closed*: This follows from the fact that $\mathbb{Z}[X]$ is a unique factorisation domain (UFD), and that any UFD is integrally closed.[1]
   - *R Noetherian*: As $\mathbb{Z}$ is Noetherian, the Hilbert basis theorem implies $\mathbb{Z}[X]$ is Noetherian.
   - *R does* not *have dimension 1*: The ideal $(X) \subset \mathbb{Z}[X]$ is a prime ideal, but the quotient $\mathbb{Z}[X]/(X) \cong \mathbb{Z}$ is only a domain and not a field, so (X) is not a maximal ideal in R.

   (b) E.g. we can take $R = \bigcup_{n \geq 1} \mathbb{C}[X^{1/n}]$.

   - *R not Noetherian:* R contains an infinite strictly ascending chain of ideals $(X) \subset (X^{1/2}) \subset (X^{1/4}) \subset \ldots$, and is thus not Noetherian (or can also note that the ideal I generated by $X^{1/n}$ over all n is not finitely generated).
   - *R has dimension 1:* Let $P \subset R$ be a nonzero prime ideal. Therefore there exists some $N \geq 1$ such that $P_N = P \cap \mathbb{C}[X^{1/N}]$ is nonzero. P_N is thus a nonzero prime ideal of the ring $\mathbb{C}[X^{1/N}]$, which implies that $\mathbb{C}[X^{1/N}]/P_N = \mathbb{C}$. Thus $P_N/C = \mathbb{C}$ and so P is a maximal ideal, as $\mathbb{C}$ is a field.
   - *R is integrally closed:* Let $K = \mathrm{Frac}R = \mathbb{C}(x^q : q \in \mathbb{Q}_{\geq 0})$. To show R integrally closed in K, first note that $\mathbb{C}[X^{1/n}]$ is a polynomial ring over a field, hence a UFD and hence integrally closed.
     Moreover, we have that the chain of inclusions $\mathbb{C}[x] \subset \mathbb{C}[x^{1/2}] \subset \mathbb{C}[x^{1/3}]$ are all integral extensions, as $x^{1/n}$ is a root of the monic equation $T^n - x$. Thus R is integrally closed.

   Another example is the ring of algebraic integers $\overline{\mathbb{Z}}$.

   (c) We can take $R = \mathbb{Z}[2i]$.

   - *R not integrally closed*: Note $\mathrm{Frac}R = \mathbb{Q}(i)$, and i is a root of the polynomial $X^2 + 1$, but $i \notin \mathbb{Z}[2i]$.

[1] For proofs of both these facts, see Theorem IV.2.3 and Proposition VII.1.7 of Lang's *Algebra* textbook.

- *R Noetherian:* We have $\mathbb{Z}[2i] \cong \mathbb{Z}[X]/(x^2+4)$, and $\mathbb{Z}[X]$ is Noetherian (by Hilbert basis theorem), and any quotient of a Noetherian domain is Noetherian.
- *R has dimension 1:* Let $P \subset \mathbb{Z}[2i]$ be a non-zero prime ideal. We have that $P \cap \mathbb{Z}$ is prime, so either $P \cap \mathbb{Z} = 0$ or $P \cap \mathbb{Z} = (p)$ for some prime p.
  We claim $P \cap \mathbb{Z}$ is nonzero. In particular, let $\alpha \in P - \{0\}$. As $\alpha \in \mathbb{Z}[2i]$, α is integral over $\mathbb{Z}$, and thus is a root of a polynomial $x^n + a_1 x^{n-1} + \cdots + a_n = 0$ for some $a_i \in \mathbb{Z}$. Therefore $a_n = -(\alpha^n + a_1\alpha^{n-1} + \cdots + \alpha a_{n-1}) \in P \cap \mathbb{Z}$.
  Thus $P \cap \mathbb{Z}$ is nonzero and so $P \cap \mathbb{Z} = (p)$. We now note the inclusion of domains $\mathbb{Z}/(p) \hookrightarrow \mathbb{Z}[2i]/P$. Thus $\mathbb{Z}[2i]/P$ is a finite domain and thus a field.

One could also take something like $\mathbb{Z}[\sqrt{5}]$ or $\mathbb{Z}[\sqrt{-3}]$. Another example is $\mathbb{C}[t^2, t^3]$.

2. (a) We first show v_f is a homomorphism. If $\alpha = f^n g/h$ and $\beta = f^m g'/h'$, then $\alpha\beta = f^{n+m}(gg')/(hh')$, and $f \nmid gg'$, $f \nmid hh'$ since f is prime. So $v_f(\alpha\beta) = n + m$. It is surjective as $v_f(f^n) = n$ for every $n \in \mathbb{Z}$.
   To see the ultrametric inequaity, let α, β be as above with $\alpha + \beta \neq 0$, and say $n \leq m$. Then
   $$\alpha + \beta = f^n \cdot \frac{gh' + f^{m-n} g'h}{hh'}.$$
   The numerator $N = gh' + f^{m-n}g'h$ is non-zero. Write $N = f^r N'$ with $r \geq 0$ and $f \nmid N'$. Since $f \nmid hh'$, we get $v_f(\alpha + \beta) = n + r \geq n = \min(v_f(\alpha), v_f(\beta))$.

   (b) We similarly show v_∞ is a homomorphism. Since k is a field, $\deg(gg') = \deg g + \deg g'$ for non-zero g, g', so $v_\infty\big(\frac{g}{h} \cdot \frac{g'}{h'}\big) = v_\infty(\frac{g}{h}) + v_\infty(\frac{g'}{h'})$. It is surjective as $v_\infty(t^{-n}) = n$ for every $n \in \mathbb{Z}$.
   To see the ultrametric inequaity, with $\alpha = g/h$, $\beta = g'/h'$ and $\alpha + \beta = (gh' + g'h)/(hh') \neq 0$, we have $\deg(gh' + g'h) \leq \max(\deg g + \deg h', \deg g' + \deg h)$, so
   $$\begin{aligned} v_\infty(\alpha + \beta) &= \deg h + \deg h' - \deg(gh' + g'h) \\ &\geq \min\big(\deg h - \deg g,\ \deg h' - \deg g'\big) = \min(v_\infty(\alpha), v_\infty(\beta)). \end{aligned}$$

   (c) Let v be a valuation on $K = k(t)$ with $v(k^\times) = 0$. We split into cases according to the sign of $v(t)$.

   - **Case 1:** $v(t) \geq 0$**.** Let $A = \{x \in K : v(x) \geq 0\}$, the valuation ring, and $\mathfrak{m} = \{x : v(x) > 0\}$, its maximal ideal. Since $k \subseteq A$ and $t \in A$, we have $k[t] \subseteq A$, so $\mathfrak{p} = \mathfrak{m} \cap k[t]$ is a prime ideal of $k[t]$.
     If $\mathfrak{p} = 0$, then $v(g) = 0$ for every non-zero $g \in k[t]$, hence $v(g/h) = 0$ for all $g/h \in K^\times$. This contradicts surjectivity of v. So $\mathfrak{p} \neq 0$, and since $k[t]$ is a PID, $\mathfrak{p} = (f)$ for a unique monic irreducible f.
     Let $c = v(f) > 0$. If $g \in k[t]$ and $f \nmid g$, then $g \in A \setminus \mathfrak{p}$, so $v(g) = 0$. Hence for $\alpha = f^n g/h$ with $f \nmid g, h$,
     $$v(\alpha) = nc + v(g) - v(h) = c\, v_f(\alpha).$$

Thus $v(K^\times) = c\mathbb{Z}$, and surjectivity forces $c = 1$. So $v = v_f$.

- **Case 2:** $v(t) < 0$**.** Put $c = -v(t) > 0$. Let $g = a_d t^d + \cdots + a_0 \in k[t]$ with $a_d \neq 0$. For each i with $a_i \neq 0$ we have $v(a_i t^i) = v(a_i) + iv(t) = -ci$, and these values are distinct for distinct i. The term $a_d t^d$ has the strictly smallest value $-cd$, so this gives $v(g) = -c \deg g$. Hence

$$v(g/h) = c(\deg h - \deg g) = c\, v_\infty(g/h),$$

and surjectivity again forces $c = 1$, so $v = v_\infty$. □

3. We shall first prove the following claim: Let $a_1, a_2, \cdots \in \mathbb{Z}_p$ be a sequence of p-adic integers. Then $\sum_{i=1}^{\infty} a_i$ converges if and only if $v(a_i) \to \infty$ as $i \to \infty$.

Proof of claim: If the series $\sum_{i=1}^{\infty} a_i$ converges, then the partial sums $S_n := \sum_{i=0}^{n} a_i$ must form a Cauchy sequence. Thus $d(s_n, s_{n=1}) \to 0$ as $n \to \infty$, this implies $v(a_{n+1}) \to \infty$ as $n \to \infty$.

Conversely, assume $v(a_i) \to \infty$ as $i \to \infty$. Then if $m > n$, then $s_m - s_n = \sum_{i=n+1}^{m} a_i$. Thus $v(s_m - s_n) \geq \min(v(a_{n+1}), \ldots, v(a_m))$, as v satisfies the ultrametric property. By our assumption, we know that, for all M, there exists N_M such that, for all $i \geq N_M$, $v(a_i) \geq M$. Hence for all $m, n \geq N(m)$, we have $v(s_m - s_n) \geq M$. This proves that s_n is Cauchy, and thus converges, as $\mathbb{Z}_p$ is complete.

To return the proving the exercise, it thus suffices to determine for which $x \in \mathbb{Z}_p$ do we have that $v(x^i/i!) \to \infty$ as $i \to \infty$.

Clearly $v(i!) \to \infty$ as $i \to \infty$. Therefore if $v(x) = 0$, then $v(x^i/i!) \not\to \infty$ as $i \to \infty$. Also, we have

$$v(i!) = \sum_{j=1}^{\infty} \left\lfloor \frac{i}{p^j} \right\rfloor \leq \sum_{j=1}^{\infty} \frac{i}{p^j} = \frac{i}{p-1}$$

by summing the geometric series. Therefore we have that

$$v(x^i/i!) = iv(x) - v(i!) \geq iv(x) - \frac{i}{p-1} = i\Big(v(x) - \frac{1}{p-1}\Big)$$

So if $v(x) > \frac{1}{p-1}$, then we have that $v(x^i/i!) \to \infty$ as $i \to \infty$, and thus the sum converges!

We must still consider the case $p = 2$. We already know that we must have $v(x) \geq 1$ and that any x such that $v(x) \geq 2$ works. We now also show that $v(x) = 1$ doesn't work.

In particular, we have

$$v(x^{2^k}/2^k!) = 2^k v(x) - (2^k - 1) = 1$$

which proves that we cannot have $v(x^i/i!) \to \infty$ as $i \to \infty$.

Therefore, the set of elements $x \in \mathbb{Z}_p$ such that the sum $\sum_{i=1}^{\infty} x^i/i!$ converges is:

$$\begin{cases} \{x \in \mathbb{Z}_p \mid v_p(x) \geq 1\} = p\mathbb{Z}_p & \text{if } p \text{ odd}, \\ \{x \in \mathbb{Z}_p \mid v_p(x) \geq 2\} = p^2\mathbb{Z}_p & \text{if } p = 2. \end{cases}$$